PROJECT DESIGN PHASE-II

DATA FLOW DIAGRAM & USER STORIES

Date:	30 September 2026	**Team ID:**	SWTID-2026-1398
Project Name:	Import Data Using Transform Maps	**Maximum Marks:**	4 Marks

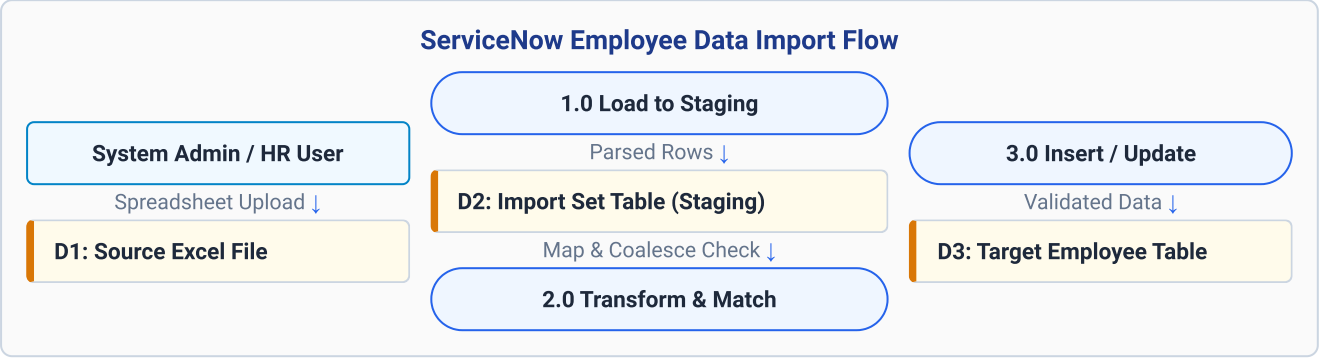
1. Project Description

This project models the automated data import process for employee records into ServiceNow utilizing **Import Sets** and **Transform Maps**. The input source is a standardized spreadsheet containing key employee metadata including *Employee ID, Name, Email, Department, and Location*.

Data Pipeline Overview: Uploaded spreadsheet data is first loaded into a temporary **Import Set Staging Table** for validation and processing. A configured **Transform Map** establishes relationships between staging columns and fields in the target Employee Table. The system dynamically utilizes **Employee ID** as the unique coalesce key to update existing records or insert new entries.

2. Data Flow Diagram (DFD) - Level 0 & Level 1

A Data Flow Diagram (DFD) visually models how information moves through the import architecture, detailing input entities, automated transformation processes, data stores, and outputs.



DFD Process Breakdown:

- **Process 1.0 (Load & Stage):** Ingests raw employee spreadsheet data into temporary Import Set staging tables.
- **Process 2.0 (Transform & Match):** Applies Transform Map field mappings and performs coalesce key matching on `Employee ID`.
- **Process 3.0 (Target Update/Insert):** Updates existing target profile records if `Employee ID` matches; otherwise inserts new employee profiles.

3. Requirements & User Stories Matrix

The following matrix details functional requirements, epics, user stories, and acceptance criteria governing the ServiceNow Import Set and Transform Map implementation.

User Type	Epic	ID	User Story / Task	Acceptance Criteria	Prio	Release
System Admin	Data Import	USN-1	As a System Admin, I want to upload an Excel file with employee details (ID, Name, Email, Dept, Location).	Data is accurately stored in the Import Set Staging Table without missing fields.	High	Sprint-1
System Admin	Field Mapping	USN-2	As a System Admin, I want to configure a Transform Map to map staging fields to the target Employee table.	All source fields map correctly to corresponding target columns in ServiceNow.	High	Sprint-1
System Admin	Deduplication	USN-3	As a System Admin, I want the system to use Employee ID as a coalesce key to detect existing records.	Existing records matching Employee ID are updated with new spreadsheet values without duplicates.	High	Sprint-1
System Admin	Record Creation	USN-4	As a System Admin, I want unmatched employee records to be inserted as new target records.	New unique Employee ID records are created successfully in the target table.	High	Sprint-1
HR Admin	Validation & Logs	USN-5	As an HR Admin, I want to view error logs for failed or invalid staging rows during execution.	Detailed import logs display specific validation or formatting error reasons.	Med	Sprint-2
HR Admin	Verification	USN-6	As an HR Admin, I want to verify that transformed employee data reflects accurately in user profiles.	Target profile cards reflect updated Email, Department, and Location values post-import.	Low	Sprint-2

4. Verification & Validation Summary

- **Coalesce Key Validation:** Verified that setting `Employee ID` as coalesce prevents duplicate profile creation in the target system.
- **Error Handling:** Rows with blank required fields or invalid format types are flagged in the Import Set log for manual review.
- **Execution Auditing:** Every transform execution creates a Transform History record detailing inserted, updated, and skipped counts.